Selected Diophantine Equations Problems

Problems 5, 7, 11, 22, 23, 27, and 33 - Detailed Solutions

Problem 5

Problem. How many lattice points (x, y) satisfy $|x| + |y| <= 7$ and $xy > 0$?

Solution.

The condition $xy > 0$ means that x and y have the same sign. Therefore, the point must lie either in Quadrant I ($x > 0, y > 0$) or in Quadrant III ($x < 0, y < 0$). Points on the axes are excluded because then $xy = 0$.

First count the points in Quadrant I. Here x and y are positive integers and

$$x + y <= 7.$$

For each possible value of x, count the possible positive values of y:

$$x = 1:\ \ y = 1, 2, ..., 6 \qquad (6 \text{ choices})$$

$$x = 2:\ \ y = 1, 2, ..., 5 \qquad (5 \text{ choices})$$

$$...$$

$$x = 6:\ \ y = 1 \qquad\qquad (1 \text{ choice})$$

Thus the number of points in Quadrant I is

$$6 + 5 + 4 + 3 + 2 + 1 = 21.$$

Every point (x,y) in Quadrant I corresponds to exactly one point $(-x,-y)$ in Quadrant III. Therefore the two quadrants contain the same number of valid points, so we double the count:

$$2 \text{ x } 21 = 42.$$

Final Answer: 42 lattice points

Problem 7

Problem. Find the number of ordered pairs (x, y) of non-negative integers satisfying 3^x + 3^y = 3^10.

Solution.

Because the equation is symmetric in x and y, assume without loss of generality that $x <= y$. Factor out 3^x:

$$3\text{^}x(1 + 3\text{^}(y-x)) = 3\text{^}10.$$

If $y > x$, then $y-x \geq 1$, so $3^{(y-x)}$ is divisible by 3. Hence

$$1 + 3^{(y-x)} \equiv 1 \pmod{3},$$

which means this factor is not divisible by 3. But the right-hand side is a pure power of 3. Therefore the factor $1 + 3^{(y-x)}$ would have to equal 1, which is impossible because it is greater than 1.

If $y = x$, then the equation becomes

$$2 \cdot 3^x = 3^{10},$$

which is also impossible because the left-hand side contains a factor of 2 while the right-hand side does not.

Thus there are no ordered pairs satisfying the equation.

Final Answer: 0 ordered pairs

Problem 11

Problem. Find the number of ordered triples (x, y, z) of positive integers satisfying $1/x + 1/y + 1/z = 1/4$, with $x \leq y \leq z$.

Solution.

Since $x \leq y \leq z$, we have $1/x \geq 1/y \geq 1/z$. Therefore

$$3/x \geq 1/x + 1/y + 1/z = 1/4,$$

so $x \leq 12$. Also, because the other two fractions are positive, $1/x < 1/4$, so $x \geq 5$. Thus

$$5 \leq x \leq 12.$$

For each fixed x, rewrite the remaining equation as

$$1/y + 1/z = 1/4 - 1/x = (x-4)/(4x).$$

For an equation $1/y + 1/z = a/b$ in lowest terms, Simon's Favorite Factoring gives

$$(ay - b)(az - b) = b^2.$$

Applying this formula to each x gives a finite divisor search. Equivalently, one may solve

$$z = 4xy / (xy - 4x - 4y)$$

and keep only integer values with $x \leq y \leq z$. The valid triples are:

(5, 21, 420)   (5, 22, 220)   (5, 24, 120)   (5, 25, 100)

(5, 28, 70)   (5, 30, 60)   (5, 36, 45)   (5, 40, 40)

(6, 13, 156)   (6, 14, 84)   (6, 15, 60)   (6, 16, 48)

$$(6, 18, 36) \quad (6, 20, 30) \quad (6, 21, 28) \quad (6, 24, 24)$$

$$(7, 10, 140) \quad (7, 12, 42) \quad (7, 14, 28) \quad (8, 9, 72)$$

$$(8, 10, 40) \quad (8, 12, 24) \quad (8, 16, 16) \quad (9, 9, 36)$$

$$(9, 12, 18) \quad (10, 10, 20) \quad (10, 12, 15) \quad (12, 12, 12)$$

There are 28 triples in this list.

Final Answer: 28 ordered triples

Problem 22

Problem. How many ordered pairs (x, y) of integers satisfy (x^2 - 4)(y^2 - 9) = 72?

Solution.

Let

$$A = x^2 - 4, \quad B = y^2 - 9.$$

Then A and B are integer divisors of 72 and AB = 72. Also, x^2 = A+4 and y^2 = B+9 must both be perfect squares.

We can sharply restrict A using congruences. Since x^2 modulo 8 can only be 0, 1, or 4,

$$A = x^2 - 4 \equiv 4, 5, \text{ or } 0 \pmod 8.$$

Likewise, since y^2 modulo 8 can only be 0, 1, or 4,

$$B = y^2 - 9 \equiv 7, 0, \text{ or } 3 \pmod 8.$$

Now check the integer factor pairs AB = 72. The positive divisors of 72 are 1, 2, 3, 4, 6, 8, 9, 12, 18, 24, 36, and 72, together with their negatives. Testing A+4 and B+9 for being perfect squares yields no admissible factor pair.

For completeness, the only divisors A for which A+4 is a nonnegative square are obtained from x^2-4 dividing 72. Checking x^2-4 ∈ {±1,±2,±3,±4,±6,±8,±9,±12,±18,±24,±36,±72} gives possible A values -4, -3, 0, 5, 12, 21, 32, 45, 60, ...; among actual divisors of 72, only A = -3 and A = 12 survive. If A = -3, then B = -24 and B+9 = -15, impossible. If A = 12, then B = 6 and B+9 = 15, not a square.

Therefore no integer values of x and y satisfy the equation.

Final Answer: 0 ordered pairs

Problem 23

Problem. Find all positive integer solutions (x, y, z) to x! + y! = z!.

Solution.

The equation is symmetric in x and y, so assume $x <= y$.

First, z cannot be less than y, because then $z! < y! < x! + y!$. Also z cannot equal y, because then the equation would give $x! = 0$. Hence $z > y$.

Divide the equation by y!:

$$x!/y! + 1 = z!/y! = (y+1)(y+2)\ldots z.$$

The right-hand side is an integer. If $x < y$, then $0 < x!/y! < 1$, so the left-hand side is not an integer, a contradiction. Therefore $x = y$.

The equation now becomes

$$2y! = z!,$$

or

$$(y+1)(y+2)\ldots z = 2.$$

Every factor on the left is at least 2. Therefore there can be only one factor, so $z = y+1$, and that factor must equal 2. Thus $y+1 = 2$, giving $y = 1$ and $z = 2$. Since $x = y$, $x = 1$.

Check: $1! + 1! = 1 + 1 = 2 = 2!$.

Final Answer: (x, y, z) = (1, 1, 2)

Problem 27

Problem. Let p be a prime. For how many primes $p < 1000$ does $x^2 + y^2 = p$ have a solution in non-negative integers (x, y)?

Solution.

By Fermat's theorem on sums of two squares, an odd prime p can be written as x^2+y^2 if and only if

$$p \equiv 1 \pmod 4.$$

The prime $p = 2$ is also representable, since $2 = 1^2 + 1^2$. Therefore we must count the primes below 1000 that are congruent to 1 modulo 4, and then include $p = 2$.

The qualifying primes are:

2, 5, 13, 17, 29, 37, 41, 53, 61, 73, 89, 97

101, 109, 113, 137, 149, 157, 173, 181, 193, 197, 229, 233

241, 257, 269, 277, 281, 293, 313, 317, 337, 349, 353, 373

389, 397, 401, 409, 421, 433, 449, 457, 461, 509, 521, 541

$$557, 569, 577, 593, 601, 613, 617, 641, 653, 661, 673, 677$$

$$701, 709, 733, 757, 761, 769, 773, 797, 809, 821, 829, 853$$

$$857, 877, 881, 929, 937, 941, 953, 977, 997$$

There are 80 primes congruent to 1 modulo 4 below 1000, plus the prime 2, for a total of 81.

Final Answer: 81 primes

Problem 33

Problem. Find all positive integer solutions to $1 + x + x^2 + x^3 = 2^y$.

Solution.

Factor the left-hand side:

$$1 + x + x^2 + x^3 = (x+1)(x^2+1).$$

Because the product is a power of 2, both positive integer factors $x+1$ and x^2+1 must themselves be powers of 2. In particular, $x+1$ is even, so x is odd.

For every odd integer x, $x^2 \equiv 1 \pmod 8$. Hence

$$x^2 + 1 \equiv 2 \pmod 8.$$

A power of 2 that is congruent to 2 modulo 8 must be exactly 2. Indeed, $2^1 = 2$, while every higher power 2^k with $k \geq 3$ is congruent to 0 modulo 8; and $2^2 = 4$.

Therefore

$$x^2 + 1 = 2,$$

so $x^2 = 1$. Since x is positive, $x = 1$. Substituting gives

$$1 + 1 + 1 + 1 = 4 = 2^2,$$

so $y = 2$.

Final Answer: $(x, y) = (1, 2)$